

Amulya Rayabhagi

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Education

The University of Texas at Dallas,

Master of Science in Computer Science (Data Science Track) | GPA: 4.00/4.00

Richardson, TX

Expected May 2028

- **Coursework:** Advanced Machine Learning, Natural Language Processing, and Big Data Analytics.

The University of Texas at Dallas,

Bachelor of Science in Computer Science | GPA: 3.75/4.00

Richardson, TX

Aug 2023 - May 2026

- **Honors:** Honors Dean's List, Collegium V Honors Student, Cum Laude Latin Honors

Technical Skills

Languages: Python (FastAPI, LangChain, asyncio), JavaScript/TypeScript (React 18, Nuxt.js, Vue), SQL, NoSQL, C++

Machine Learning: Retrieval-Augmented Generation (RAG), Large Language Models (LLMs), Multi-Agent Orchestration, YOLOv8, CNNs, XGBoost

Tools & Infrastructures: ChromaDB, Redis, MongoDB, SQLite, Docker, Git, Prisma ORM, Better Auth, RESTAPI, Vitest

Certifications: Docker Foundations Professional Certificate – Docker Inc. (2026)

Work Experience

Software Engineering Intern

Aug 2025 - Dec 2025

LLM Software

Dallas, TX

- Accelerated financial claim processing times by 25% by building a high-performance web platform using FastAPI and React integrated with predictive **XGBoost machine learning models** to streamline data workflows and reduce manual review delays.
- Attained a 95% classification precision rate for complex financial data by implementing specialized **multi-agent AI** workflows that run automated validation checks to guarantee high data accuracy and minimize processing errors.
- Optimized **GenAI** system efficiency and server performance by designing persistent memory structures for multistep data validations, significantly reducing server workload and ensuring continuous processing of complex financial tasks.

ERP Security Administrator

May 2025 - Aug 2025

The University of Texas at Dallas

Richardson, TX

- Engineered robust **Oracle PeopleSoft** security objects, permission lists, and multi-tier access control structures across HCM, FMS, and Campus Solutions modules for over 40,000 campus users by analyzing entry points, troubleshooting production issues using **PeopleTools**, and adhering to strict SDLC standards to ensure 100% compliance with University **Identity and Access Management (IAM)** policies and change management controls.
- Hardened application zone integrity and streamlined user provisioning across 4 distinct system environments by deploying federated authentication frameworks and designing automated, repeatable security workflows to eliminate unauthorized access vulnerabilities and reduce manual provisioning processing times by 30%.
- Resolved over 150 user-facing access incidents across Fluid UI portals and achieved a 95% first-contact resolution rate while drastically reducing ticketing latency by utilizing the Application Engine to test/debug access anomalies and developing standardized technical change management documentation.

Projects

Guidepoint: Mobile Navigation App for Visually Impaired

Jan 2026 - May 2026

- Achieved a 0mm tracking divergence for an indoor navigation app by combining data from **IMU sensors** via **React Native**, completely eliminating location drift and maintaining perfect positioning without GPS.
- Created an AI-powered pedestrian safety system by training a custom **CNN** (Convolutional Neural Network) and **YOLOv8** framework using **Python** to detect street hazards which successfully protected users by accurately classifying dangers with a 92% success rate.
- Reduced mobile device processing strain by mapping an optimized A* pathfinding algorithm against geometric floor grids to cut real-time navigation and routing delays by 25% for a smoother user experience.

Elevatr: AI Web Platform for Distributed Career Analytics

Jan 2026 - Apr 2026

- Scaled document processing execution to handle 500+ tokens per second concurrently, eliminating document parsing latency by designing an asynchronous **Retrieval-Augmented Generation (RAG)** web pipeline via **LangChain** and **Python** asyncio.
- Boosted textual profile context matching accuracy to 94.2% and mapped instant skill gaps by extracting raw applicant datasets from MongoDB, generating dense vector embeddings, and indexing them into a semantic **ChromaDB** vector database.
- Cut rendering latency by 35% under concurrent requests while optimizing platform loading times and state efficiency by building a dual-persistence database matrix pairing **MongoDB** with **Redis** caching via TanStack Query.

RainbowRoundup: Events Calendar PWA

Aug 2024 - Dec 2024

- Managed secure multi-tier RBAC access control structures across 4 distinct user roles and deployed a centralized DFW regional operations dashboard onto **AWS EC2** by developing a full-stack **Progressive Web App (PWA)** using **Nuxt.js** and **Prisma ORM**.
- Decreased core database session overhead by 65% by deploying a securely signed, 5-minute client-side cookie validation algorithm alongside an Email OTP authentication configuration via the **Better Auth** architecture module.
- Engineered a high-concurrency event tracking environment to support scalable, real-time push alerts by combining live programmatic Google Places and Google Calendar API integrations, asynchronous scheduled email/native automation, and automated tracking modules containerized via **Docker**.